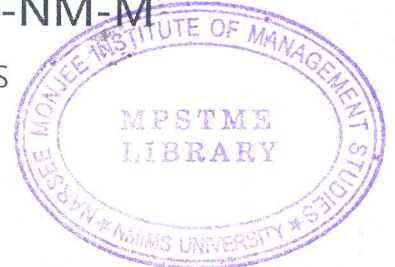


SVKMs NMIMS
MUKESH PATEL SCH.OF TECH.MGMT ENGG-NM-M

Programme: MASTER OF COMPUTER APPLICATIONS

Year: I/Semester II (Exam Year: 2023-2024)



Subject: Software Engineering

Date: 17 Apr 2024

Time: 10:00 am - 01:00 pm (03:00 Hrs.)

Max Marks: 100

FINAL EXAMINATION(2023-2024)

Instructions:

1. This question paper contains 2 pages
2. Answer to each new question to be started on a fresh page.
3. Figure in right hand side indicates full marks
4. Question no.1 is compulsory
5. Assume Suitable data if necessary

- | | |
|--|----|
| 1. Answer briefly | 20 |
| a. Define Software Engineering. Describe the types of software in detail. | 5 |
| b. Describe the Importance of UML in the context of software development. | 5 |
| c. List the symbols used in a data flow diagram and explain their meanings. | 5 |
| d. Summarize the importance of design quality in the development of software. | 5 |
| 2. Answer the following questions(Attempt Any 4 Questions) | 80 |
| 1. Answer the following questions | 20 |
| a. Describe how extreme programming (XP) methodology can be applied for a software project development? | 10 |
| b. Describe the key components of a sequence diagram and their purposes with a neat labelled diagram for blog posting. | 10 |
| 2. Answer the following questions | 20 |
| a. Compare and contrast Functional and Non Functional requirements in software development. | 10 |

- b. Explain how does Software engineering ensure that a complex system meets its operational requirements? 10
3. Answer the following questions 20
- a. Implement use-case diagram for below case study. A group diary and time management system is intended to support the timetabling of meetings and appointments across a group of co-workers. When an appointment is to be made that involves a number of people, the system finds a common slot in each of their diaries and arranges the appointment for that time. If no common slots are available, it interacts with the user to rearrange his or her personal diary to make room for the appointment. 5
- b. Explain the significance of Interface Design Principles and Guidelines in the development of a Web Application. 5
- c. What is the Concurrent Development Model, and how does it differ from other software development models? 10
4. Answer the following questions 20
- a. Describe the primary requirements of user interface design. 5
- b. Compare and contrast White Box Testing and Black Box Testing. 5
- c. Describe activity diagram for library management system with a neat labelled diagram and explain the components? 10
5. Answer the following questions 20
- a. What is the primary purpose of unit testing, and how does it contribute to software quality? 5
- b. Determine the most suitable behavior modeling approach for a given software system based on its requirements and constraints. 10
6. Answer the following questions 20
- a. What are the key characteristics of the layered architectural style? 5
- b. Determine which is better McCall's Software Quality Factors or ISO 9126 Quality Factors. 5
- c. Summarize agile process and extreme programming. 10